

Multimodelling Developement System

UNIT -1

Contents -

- Overview of multimodelling and model- driven engineering
- Types of models: continuous, discrete, stochastic, hybrid
- Benefits and challenges of multimodelling
- Applications in engineering, AI, and simulation systems

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Model -

- Models are abstract, mathematical, or logical representations of systems, data, or processes used to simulate, analyze, and predict behavior.
- They simplify real-world complexity to facilitate understanding and design, spanning areas like machine learning, databases, software engineering, and theoretical computation.

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Model -

Key types of models include:

- **Machine Learning/AI Models:** Algorithms trained on data to identify patterns and make predictions or decisions.
- **Data Models:** Conceptual tools that define the structure, storage, and relationships of data in a database.

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Models -

Key types of models include:

Computational/Simulation Models: Digital replicas of systems (e.g., weather, traffic) used to test "what-if" scenarios.

Theoretical Models (Computation): Abstract frameworks like Turing machines used to measure algorithm complexity and capability.

Software/System Models: Blueprints for software development or architecture (e.g., Waterfall model, MVC).

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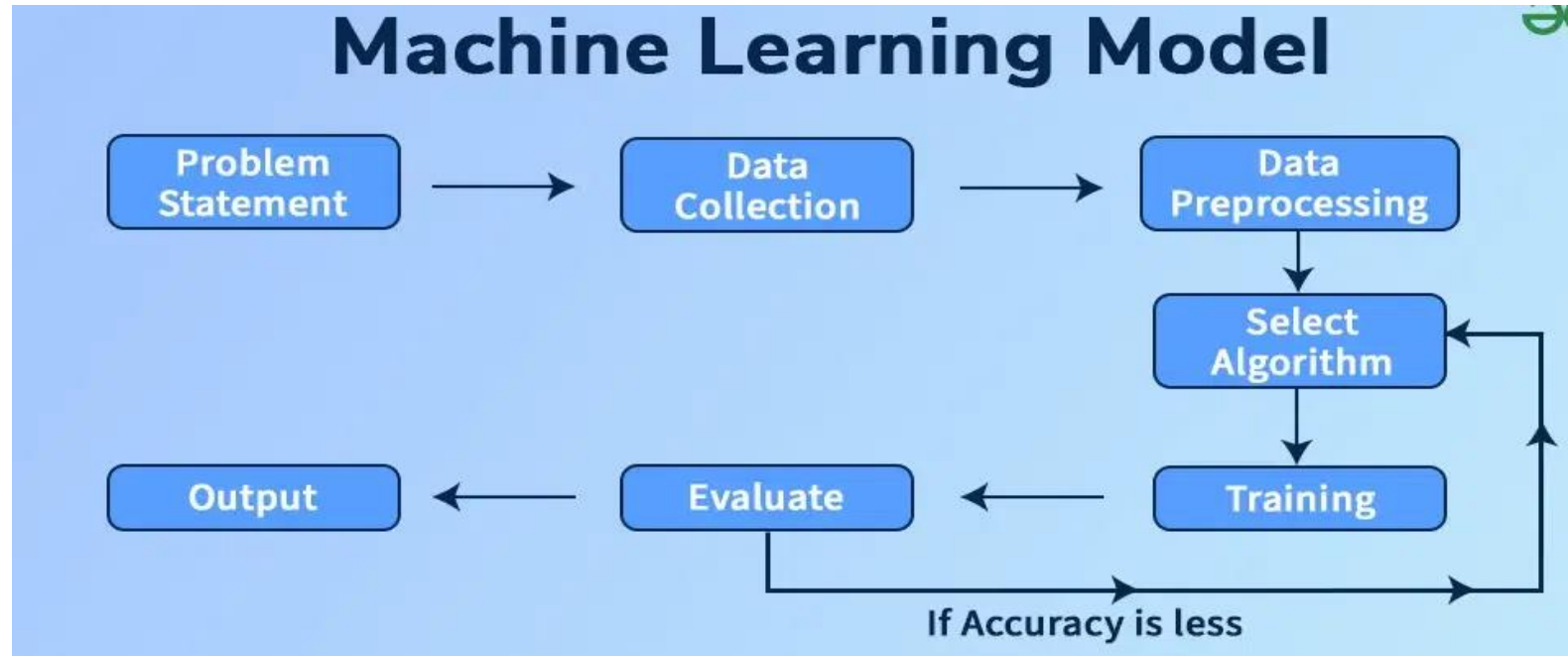
Models -

Purposes of Modeling:

- **Prediction:** Forecasting future outcomes based on historical data.
- **Abstraction:** Reducing complex systems into manageable components.
- **Simulation:** Testing, exploring, or verifying systems without needing the physical entity.

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Models - Example



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Models -

A Machine Learning Model is a computational program that learns patterns from data and makes decisions or predictions on new, unseen data. It is created by training a machine learning algorithm on a dataset and optimizing it to minimize errors. Key characteristics of ML models are:

- Finds hidden patterns from historical information.
- Can forecast values or classify inputs.
- Learns from additional data and feedback.
- Reduces human effort and increases efficiency.

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- Multimodelling is a computational approach that combines different types of models to simulate and design complex systems, particularly Cyber-Physical Systems and Systems-of-Systems.
- It is closely related to Model-Driven Engineering (MDE), an approach where models, rather than code, are the primary artifacts used to raise the level of abstraction and automate software development.

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- Multimodelling involves integrating models from different domains and disciplines (e.g., electrical, mechanical, human behavior) to provide a more comprehensive representation of a system than a single model could achieve alone.
- This often involves co-simulation, where various specialized simulation tools work together.
- Model-Driven Engineering (MDE) focuses on the systematic use of models throughout the software development lifecycle. Key aspects include:

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- **Abstraction:** Considering systems at higher levels of abstraction, focusing on domain logic rather than implementation details.
- **Automation:** Using model transformations (e.g., Platform-Independent Models to Platform-Specific Models) and code generators to automate tasks such as validation, testing, and code production.
- **Standards:** Often relying on standards like the OMG's Model-Driven Architecture (MDA) and languages such as UML or Domain-Specific Languages (DSLs).

Types of Models

Multimodelling often integrates different types of models, classified based on how they represent time and uncertainty:

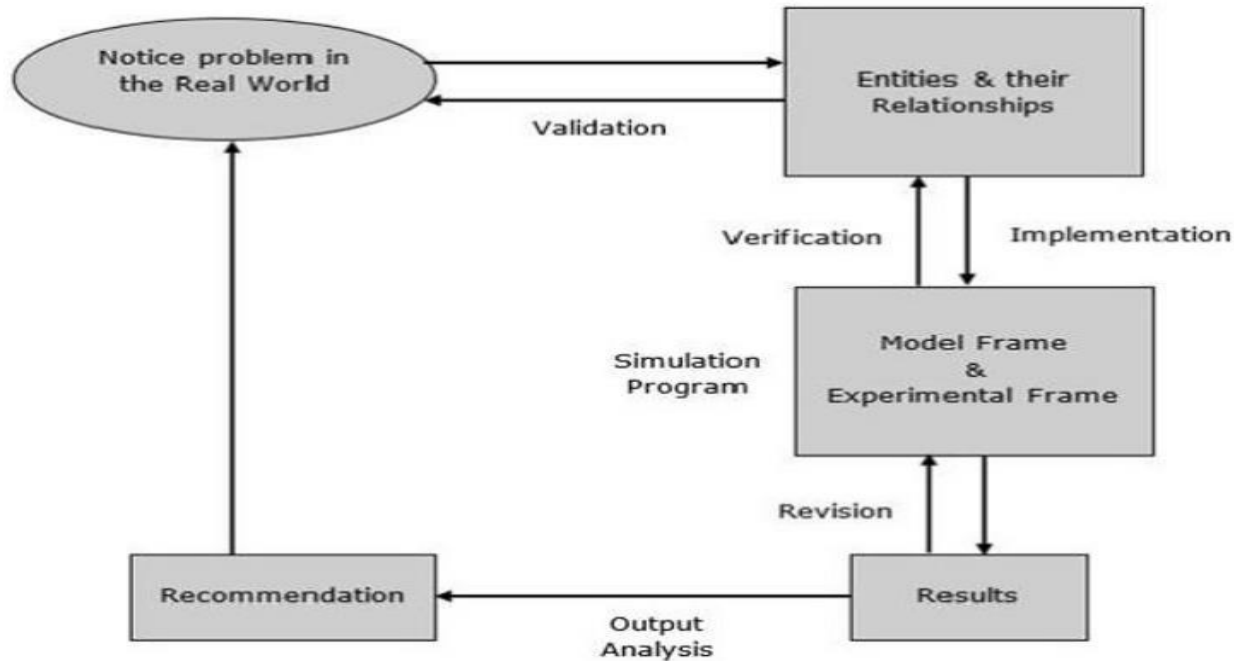
- **Continuous Models:** These models describe systems where states change continuously over time, often using differential equations. They are typically used for physical processes like heat transfer or fluid dynamics.
- **Discrete Models:** In these models, system states change only at a countable set of points in time (discrete events). They are commonly used in manufacturing process control or computer network simulations.

Types of Models

Multimodelling often integrates different types of models, classified based on how they represent time and uncertainty:

- **Stochastic Models:** These models incorporate randomness and probability distributions to account for uncertainty in a system's behavior. They are used in finance, epidemic modeling, and performance analysis where outcomes are not fully predictable.
- **Hybrid Models:** These models combine two or more of the other types to capture different aspects of a system. For example, a hybrid model might combine continuous dynamics with discrete event logic or integrate mechanistic modeling with machine learning techniques.

Modelling Process



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Step 1 – Examine the problem. In this stage, we must understand the problem and choose its classification accordingly, such as deterministic or stochastic.

Step 2 – Design a model. In this stage, we have to perform the following simple tasks which help us design a model –

- Collect data as per the system behavior and future requirements.
- Analyze the system features, its assumptions and necessary actions to be taken to make the model successful.

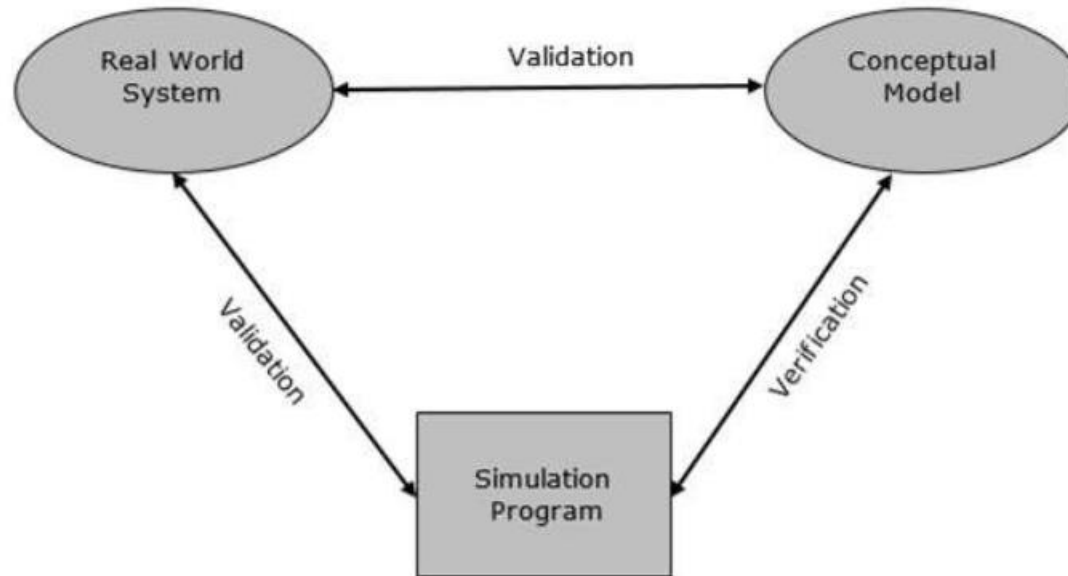
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- Determine the variable names, functions, its units, relationships, and their applications used in the model.
- Solve the model using a suitable technique and verify the result using verification methods. Next, validate the result.
- Prepare a report which includes results, interpretations, conclusion, and suggestions.

Step 3 – Provide recommendations after completing the entire process related to the model. It includes investment, resources, algorithms, techniques, etc.

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Verification & Validation



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Verification & Validation

- **Validation** is the process of comparing two results. In this process, we need to compare the representation of a conceptual model to the real system. If the comparison is true, then it is valid, else invalid.
- **Verification** is the process of comparing two or more results to ensure its accuracy. In this process, we have to compare the models implementation and its associated data with the developer's conceptual description and specifications.

Verification & Validation Techniques

Techniques to Perform Verification of Simulation Model

Following are the ways to perform verification of simulation model –

- ✓ By using programming skills to write and debug the program in sub-programs.
- ✓ By using Structured Walk-through policy in which more than one person is to read the program.
- ✓ By tracing the intermediate results and comparing them with observed outcomes.
- ✓ By checking the simulation model output using various input combinations.
- ✓ By comparing final simulation result with analytic results.

Verification & Validation Techniques

Techniques to Perform Validation of Simulation Model

Step 1 – Design a model with high validity. This can be achieved using the following steps

- ✓ The model must be discussed with the system experts while designing.
- ✓ The model must interact with the client throughout the process.
- ✓ The output must supervised by system experts.

Step 2 – Test the model at assumptions data. This can be achieved by applying the assumption data into the model and testing it quantitatively. Sensitive analysis can also be performed to observe the effect of change in the result when significant changes are made in the input data.

Verification & Validation Techniques

Step 3 – Determine the representative output of the Simulation model. This can be achieved using the following steps –

- ✓ Determine how close is the simulation output with the real system output.
- ✓ Comparison can be performed using the Turing Test. It presents the data in the system format, which can be explained by experts only.
- ✓ Statistical method can be used for compare the model output with the real system output.

Benefits and Challenges of Multimodelling

Benefits

Higher Abstraction: Allows engineers to focus on the problem domain rather than low-level code, improving communication across teams.

Increased Productivity: Automated code generation can significantly reduce manual coding effort and accelerate development time.

Improved Quality and Consistency: Automatically generated code is typically more consistent and less prone to human error/bugs than hand-written code.

Easier Adaptation to New Platforms: Platform-Independent Models can be re-targeted to new technologies with new transformations, rather than rewriting the entire system.

Challenges

Initial Investment: Requires significant initial investment in tools, training, and developing translators/transformers for new platforms.

Complexity Management: Integrating diverse models (often with different tools and data formats) can be complex and challenging to manage.

Validation and Verification: Ensuring the accuracy and reliability of the integrated multi-model system can be more difficult than with single models.

Misunderstandings/Discrepancies: Differences between models and actual system properties can arise due to scaling effects, material properties, or domain misunderstandings.

Applications in Engineering, AI, and Simulation

Multimodelling is essential for developing complex, interconnected systems:

❖ **Engineering -**

It is extensively used in the design of cyber-physical systems, such as automotive and aerospace systems, where mechanical, electrical, and software components must work in unison

- ✓ **Cyber-Physical Systems (CPS):** Used to design and simulate systems that integrate computation, networking, and physical processes, such as autonomous vehicles and smart manufacturing.

❖ ?Simulation Systems:

Multimodelling allows for the creation of sophisticated simulations for complex scenarios, such as urban planning, climate change forecasting, and logistical supply chains, by combining different simulation paradigms .

- ✓ **Multi-physics Simulations:** Essential for designing electric machines or aircraft where electromagnetic, thermal, and structural aspects are interrelated and must be modeled together.
- ✓ **Digital Twins:** Creation of virtual replicas of physical systems that use integrated models and real-time data for monitoring, simulation, and predictive maintenance.

❖ Artificial Intelligence (AI):

In AI, the term multimodelling often refers to multimodal AI, which integrates data from various sources (e.g., text, images, audio, video) to create more robust and capable AI systems, enabling applications like image captioning and speech recognition .

- ✓ **AI-Enhanced Simulations:** AI models (like neural networks) are trained on simulation data to accelerate computationally intensive tasks (e.g., CFD analysis) and make simulations faster and more accessible to non-experts.
- ✓ **Hybrid AI/Mechanistic Models:** Combining traditional, physics-based models with machine learning algorithms for improved accuracy in areas like flood prediction or materials science.
- ✓ **Model Validation:** AI techniques like deep reinforcement learning can be used to validate models and generate optimal control mechanisms in simulation environments.

Modelling & Simulation

- **Modelling** is the process of representing a model which includes its construction and working. This model is similar to a real system, which helps the analyst predict the effect of changes to the system. In other words, modelling is creating a model which represents a system including their properties. It is an act of building a model.
- **Simulation** of a system is the operation of a model in terms of time or space, which helps analyze the performance of an existing or a proposed system. In other words, simulation is the process of using a model to study the performance of a system. It is an act of using a model for simulation.

History of Simulation

The historical perspective of simulation is as enumerated in a chronological order -

1940 – A method named Monte Carlo was developed by researchers (John von Neumann, Stanislaw Ulan, Edward Teller, Herman Kahn) and physicists working on a Manhattan project to study neutron scattering.

1960 – The first special-purpose simulation languages were developed, such as SIMSCRIPT by Harry Markowitz at the RAND Corporation.

1970 – During this period, research was initiated on mathematical foundations of simulation.

1980 – During this period, PC-based simulation software, graphical user interfaces and object-oriented programming were developed.

1990 – During this period, web-based simulation, fancy animated graphics, simulation-based optimization, Markov-chain Monte Carlo methods were developed.

Developing Simulation Models

Simulation models consist of the following components: system entities, input variables, performance measures, and functional relationships. Following are the steps to develop a simulation model.

Step 1 – Identify the problem with an existing system or set requirements of a proposed system.

Step 2 – Design the problem while taking care of the existing system factors and limitations.

Step 3 – Collect and start processing the system data, observing its performance and result.

Step 4 – Develop the model using network diagrams and verify it using various verifications techniques.

Step 5 – Validate the model by comparing its performance under various conditions with the real system.

Developing Simulation Models

Step 6 – Create a document of the model for future use, which includes objectives, assumptions, input variables and performance in detail.

Step 7 – Select an appropriate experimental design as per requirement.

Step 8 – Induce experimental conditions on the model and observe the result.

Performing Simulation Analysis

Following are the steps to perform simulation analysis.

Step 1 – Prepare a problem statement.

Step 2 – Choose input variables and create entities for the simulation process. There are two types of variables - decision variables and uncontrollable variables. Decision variables are controlled by the programmer, whereas uncontrollable variables are the random variables.

Step 3 – Create constraints on the decision variables by assigning it to the simulation process.

Step 4 – Determine the output variables.

Step 5 – Collect data from the real-life system to input into the simulation.

Step 6 – Develop a flowchart showing the progress of the simulation process.

Performing Simulation Analysis

Step 7 – Choose an appropriate simulation software to run the model.

Step 8 – Verify the simulation model by comparing its result with the real-time system.

Step 9 – Perform an experiment on the model by changing the variable values to find the best solution.

Step 10 – Finally, apply these results into the real-time system.

Modelling & Simulation Advantages

Following are the advantages of using Modelling and Simulation –

- **Easy to understand** – Allows to understand how the system really operates without working on real-time systems.
- **Easy to test** – Allows to make changes into the system and their effect on the output without working on real-time systems.
- **Easy to upgrade** – Allows to determine the system requirements by applying different configurations.

Modelling & Simulation Advantages

Following are the advantages of using Modelling and Simulation –

- **Easy to identifying constraints** – Allows to perform bottleneck analysis that causes delay in the work process, information, etc.
- **Easy to diagnose problems** – Certain systems are so complex that it is not easy to understand their interaction at a time. However, Modelling & Simulation allows to understand all the interactions and analyze their effect. Additionally, new policies, operations, and procedures can be explored without affecting the real system.

Modelling & Simulation Disadvantages

Following are the disadvantages of using Modelling and Simulation –

- ✓ Designing a model is an art which requires domain knowledge, training and experience.
- ✓ Operations are performed on the system using random number, hence difficult to predict the result.
- ✓ Simulation requires manpower and it is a time-consuming process.
- ✓ Simulation results are difficult to translate. It requires experts to understand.
- ✓ Simulation process is expensive.

Modelling & Simulation Application Areas

Modelling & Simulation can be applied to the following areas – Military applications, training & support, designing semiconductors, telecommunications, civil engineering designs & presentations, and E-business models.

Additionally, it is used to study the internal structure of a complex system such as the biological system. It is used while optimizing the system design such as routing algorithm, assembly line, etc. It is used to test new designs and policies. It is used to verify analytic solutions.

Models & Events

Following are the basic concepts of Modelling & Simulation.

- ✓ **Object** is an entity which exists in the real world to study the behavior of a model.
- ✓ **Base Model** is a hypothetical explanation of object properties and its behavior, which is valid across the model.
- ✓ **System** is the articulate object under definite conditions, which exists in the real world.
- ✓ **Experimental Frame** is used to study a system in the real world, such as experimental conditions, aspects, objectives, etc. Basic Experimental Frame consists of two sets of variables – the Frame Input Variables & the Frame Output Variables, which matches the system or model terminals. The Frame input variable is responsible for matching the inputs applied to the system or a model. The Frame output variable is responsible for matching the output values to the system or a model.

Models & Events

- ✓ **Lumped Model** is an exact explanation of a system which follows the specified conditions of a given Experimental Frame.
- ✓ **Verification** is the process of comparing two or more items to ensure their accuracy. In Modelling & Simulation, verification can be done by comparing the consistency of a simulation program and the lumped model to ensure their performance.
- ✓ **Validation** is the process of comparing two results. In Modelling & Simulation, validation is performed by comparing experiment measurements with the simulation results within the context of an Experimental Frame. The model is invalid, if the results mismatch. There are various ways to perform validation process

System State Variables

The system state variables are a set of data, required to define the internal process within the system at a given point of time.

In a discrete-event model, the system state variables remain constant over intervals of time and the values change at defined points called event times.

In continuous-event model, the system state variables are defined by differential equation results whose value changes continuously over time.

Following are some of the system state variables –

- ✓ **Entities & Attributes** – An entity represents an object whose value can be static or dynamic, depending upon the process with other entities. Attributes are the local values used by the entity.
- ✓ **Resources** – A resource is an entity that provides service to one or more dynamic entities at a time. The dynamic entity can request one or more units of a resource; if accepted then the entity can use the resource and release when completed. If rejected, the entity can join a queue.
- ✓ **Lists** – Lists are used to represent the queues used by the entities and resources. There are various possibilities of queues such as LIFO, FIFO, etc. depending upon the process.
- ✓ **Delay** – It is an indefinite duration that is caused by some combination of system conditions.



Types of simulation Models

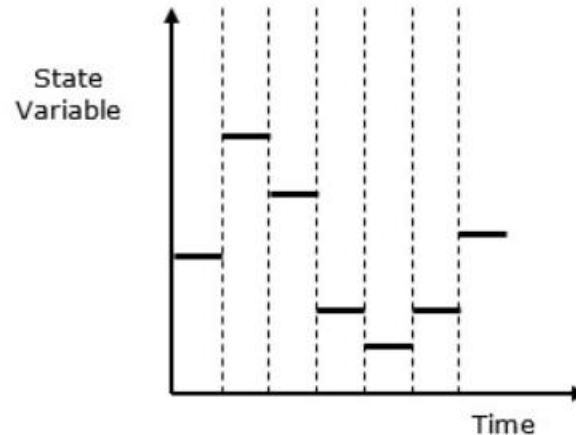
Types of simulation Models

A system can be classified into the following categories.

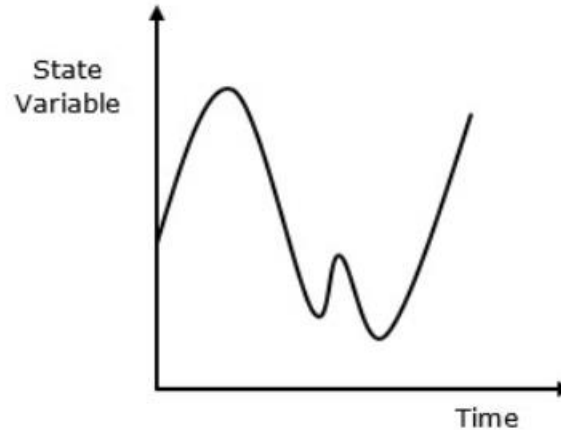
- **Discrete-Event Simulation Model** – In this model, the state variable values change only at some discrete points in time where the events occur. Events will only occur at the defined activity time and delays.
- **Stochastic vs. Deterministic Systems** – Stochastic systems are not affected by randomness and their output is not a random variable, whereas deterministic systems are affected by randomness and their output is a random variable.
- **Static vs. Dynamic Simulation** – Static simulation include models which are not affected with time. For example: Monte Carlo Model. Dynamic Simulation include models which are affected with time.

Types of simulation Models

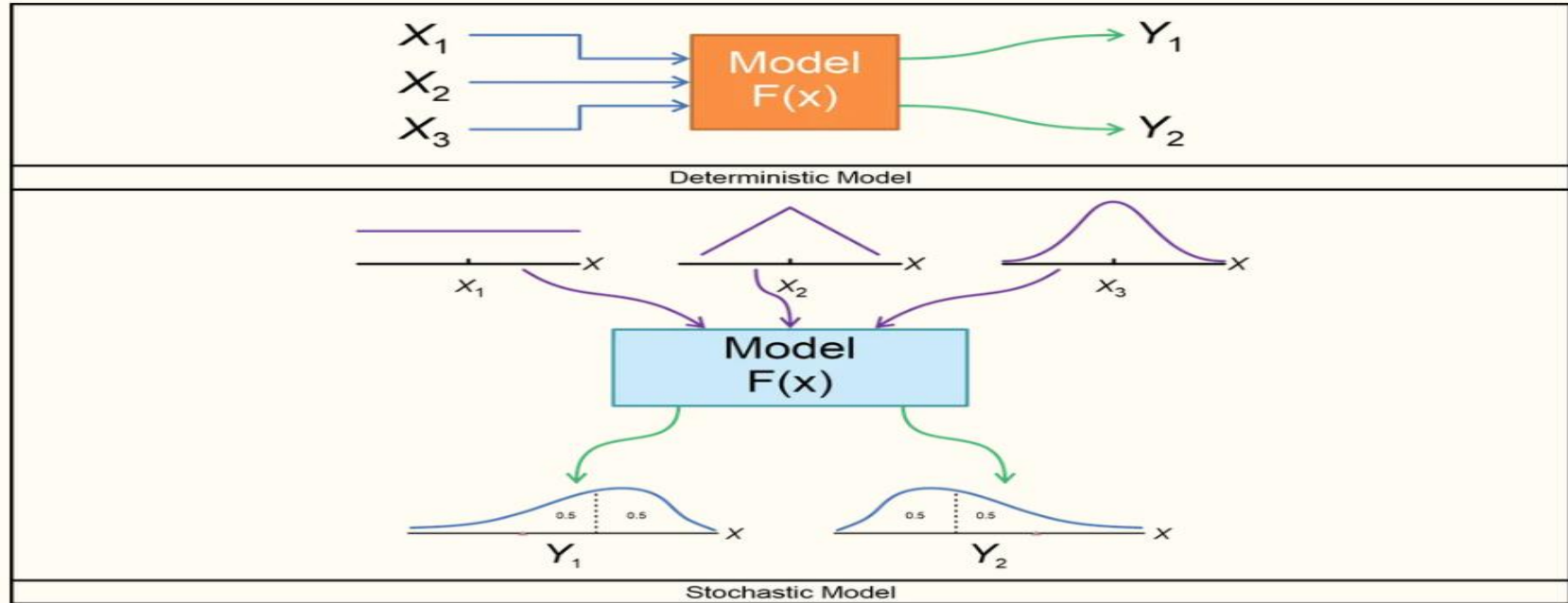
Discrete vs. Continuous Systems – Discrete system is affected by the state variable changes at a discrete point of time. Its behavior is depicted in the following graphical representation.



Continuous system is affected by the state variable, which changes continuously as a function with time. Its behavior is depicted in the following graphical representation.



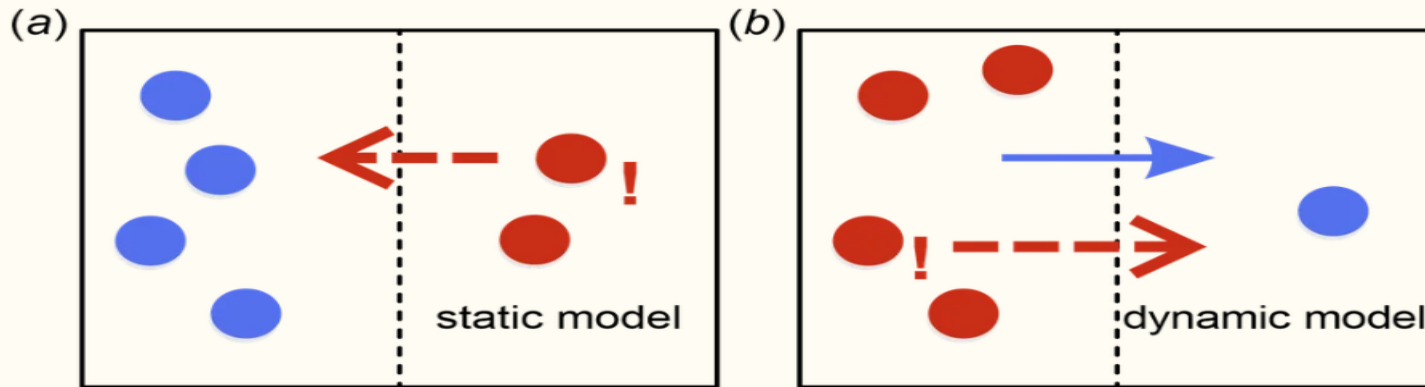
Deterministic vs. stochastic models



Key points – deterministic models are used when the outcomes can be fully predicted, while stochastic models are used when the variables in the process are unknown.

- **Deterministic models:** Applied when dealing with processes whose behavior can be fully predicted from start to end. With a given set of inputs, deterministic models always produce the same outputs. There is no randomness in deterministic models since we know the precise values of all input variables.
- **Stochastic models:** Applied whenever we cannot accurately estimate input variables – either because we don't have enough information about the variables or because they fluctuate within a specific range.

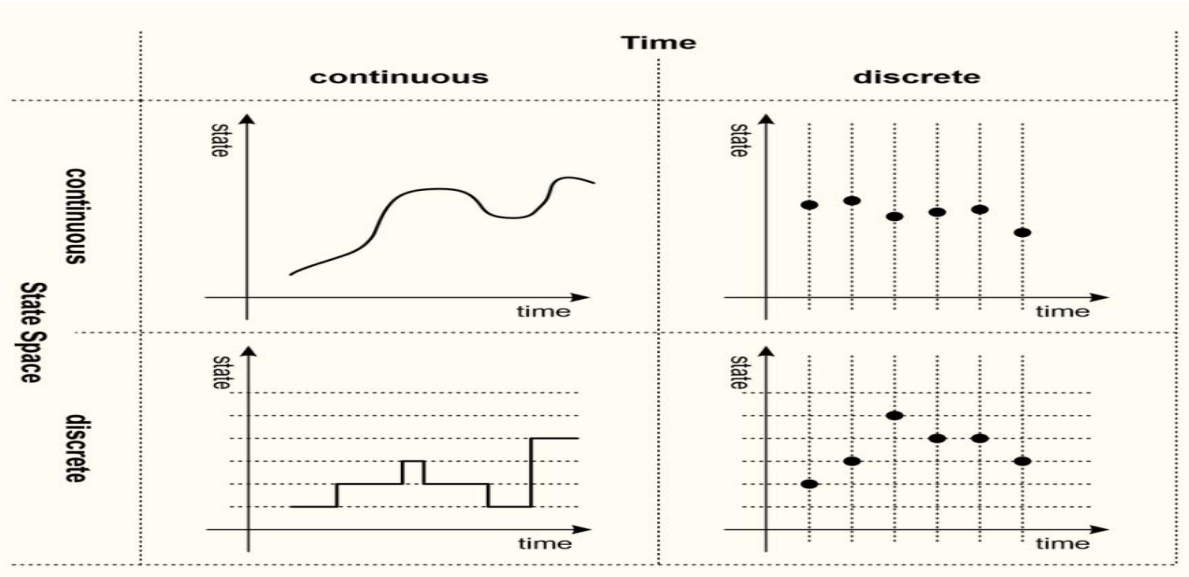
Static vs. dynamic models



Key points – static models describe a process at one point in time, while dynamic models can represent processes that change and develop over time.

- **Static models:** Applied to simulate processes at a specific instant in time. The outputs of these models only depend on the inputs and internal model variables. Static models produce a snapshot of the process in a specified timeframe.
- **Dynamic models:** Take into account not only inputs at the current time, but also outputs at previous points in time. Additionally, the output of dynamic models depends on time itself (i.e., variables in dynamic models are denoted as functions of time).

Discrete vs. continuous models



Key points – discrete models are used with systems that change at specific, countable points in time. Continuous models are designed to deal with processes that change continuously.

- **Discrete models:** Applied when the specific instant at which an event occurs, with no changes occurring between events, can be pinpointed. Examples of discrete processes are app downloads, product purchases, or customer arrivals at a bank.
- ✓ In discrete systems, the changes in the system state are discontinuous and each change in the state of the system is called an event. The model used in a discrete system simulation has a set of numbers to represent the state of the system, called as a state descriptor.
- **Continuous models:** Applied to processes whose state changes continuously. Examples of continuous processes and phenomena are temperature, speed, or revenue.

In discrete event simulation along with simulation of time-sharing system.

Simulation of Time-Sharing System -

Time-sharing system is designed in such a manner that each user uses a small portion of time shared on a system, which results in multiple users sharing the system simultaneously. The switching of each user is so rapid that each user feels like using their own system. It is based on the concept of CPU scheduling and multi-programming where multiple resources can be utilized effectively by executing multiple jobs simultaneously on a system.

Example – SimOS Simulation System.

It is designed by Stanford University to study the complex computer hardware designs, to analyze application performance, and to study the operating systems. SimOS contains software simulation of all the hardware components of the modern computer systems, i.e. processors, Memory Management Units (MMU), caches, etc.

Monte Carlo Simulation

- ✓ Monte Carlo simulation is a computerized mathematical technique to generate random sample data based on some known distribution for numerical experiments.
- ✓ This method is applied to risk quantitative analysis and decision making problems. This method is used by the professionals of various profiles such as finance, project management, energy, manufacturing, engineering, research & development, insurance, oil & gas, transportation, etc.
- ✓ This method was first used by scientists working on the atom bomb in 1940. This method can be used in those situations where we need to make an estimate and uncertain decisions such as weather forecast predictions.

Monte Carlo Simulation Important Characteristics -

Following are the three important characteristics of Monte-Carlo method –

- ✓ Its output must generate random samples.
- ✓ Its input distribution must be known.
- ✓ Its result must be known while performing an experiment.

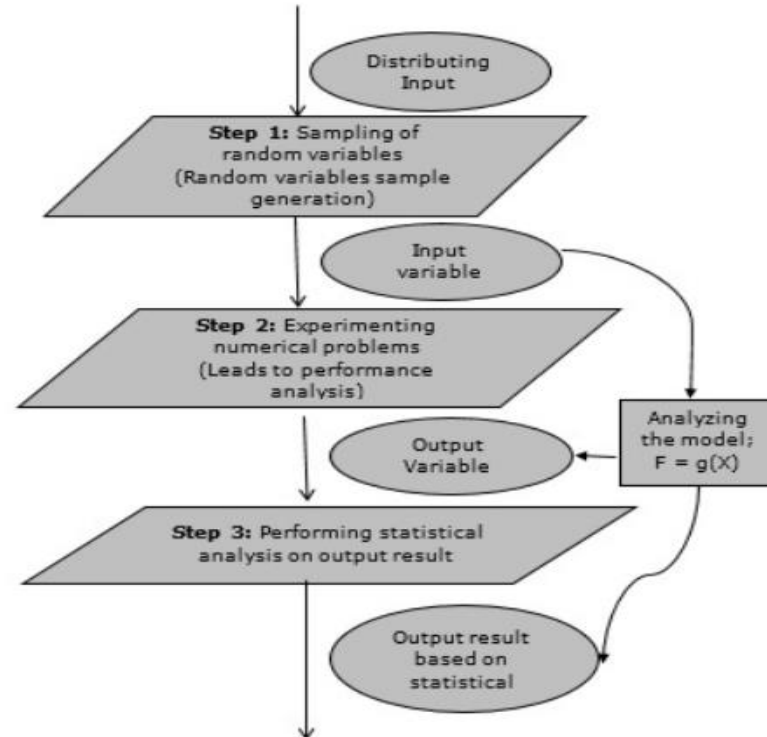
Monte Carlo Simulation Advantages

- ✓ Easy to implement.
- ✓ Provides statistical sampling for numerical experiments using the computer.
- ✓ Provides approximate solution to mathematical problems.
- ✓ Can be used for both stochastic and deterministic problems.

Monte Carlo Simulation Disadvantages

- ✓ Time consuming as there is a need to generate large number of sampling to get the desired output.
- ✓ The results of this method are only the approximation of true values, not the exact.

Monte Carlo Simulation Method Flow Diagram





Thank You